Band-1 Amicable Pairs Probe: Opposite-Parity / Coprime Existence (Collapse 2.28)

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Band-1 Amicable Pairs: Finite Catalog Probe + Open Existence Questions

Honesty stamp. This note enumerates amicable pairs with smaller member $\leq 10^6$ on the box and records the **open** status of opposite-parity and coprime existence. It does **not** claim infinitude of amicable pairs. It does **not** claim a Clay-millennial or ToE result. **GATE-04** is locked out of scope.

> **Opposite-parity amicable pairs: OPEN.**
> **Coprime amicable pairs: OPEN.**
> **Infinitude of amicable pairs: OPEN — not claimed.**

Abstract

An amicable pair (a, b) with $a < b$ satisfies $\sigma(a) = \sigma(b) = a + b$, where σ is the sum-of-divisors function.

Box result. Linear sieve for σ up to $3 \cdot 10^6$ yields **exactly 42** amicable pairs with smaller member $\leq 10^6$, matching OEIS A066873 ($a(6) = 42$; also $a(5) = 13$ for $< 10^5$). Among these 42:

Property	Count in bound
Opposite parity (even–odd)	**0**
Coprime (gcd = 1)	**0**
Both even	**34**
Both odd	**8**

Literature. Existence of an opposite-parity pair and of a coprime pair remain **open**. D'Abramo (arXiv math/0501402) proves: if an opposite-parity pair exists, the odd member is a perfect square and the even member is 2^k times an odd perfect square. Exhaustive catalogs (Chernykh / BOINC) report no opposite-parity or coprime examples through enormous bounds; that is empirical evidence, **not** a proof of nonexistence. Infinitude remains open despite $> 10^9$ known pairs.

Problem (Band-1 collapse item 2.28)

1. Cite current status: catalogs; opposite-parity / coprime open questions.
2. Enumerate all amicable pairs with smaller member $\leq B$ (here $B = 10^6$) and check parity / gcd.
3. Keep infinitude, GATE-04, and Clay rhetoric off the claim surface.

Literature ledger (2024–2026 web check; not re-proved on box)

Topic	Status	Source
Count smaller $< 10^n$	OEIS A066873: $a(5) = 13$, $a(6) = 42$, … $a(20) = 4197267$	OEIS
Exhaustive catalog	All pairs with smaller $< 10^{20}$; BOINC finished 10^{21} (Apr 2026; ~4.8M twenty-one-digit amicables)	Chernykh https://sech.me/ap/ ; BOINC forum
Opposite parity	**OPEN**; conditional form if exists	D'Abramo math/0501402; formal-conjectures #2241
Coprime	**OPEN**; all known pairs have gcd > 1	formal-conjectures #2244; catalogs
Infinitude	**OPEN**	standard; Chen–Tang–Zhan arXiv:2601.07444 (Lean formalization of statements)
Odd–odd pairs	Exist in catalogs (first: $(12285, 14595)$)	OEIS / classical

Canonical conditional (opposite parity).
G. D'Abramo, *On Amicable Numbers With Different Parity*, arXiv:math/0501402.
If (M, N) amicable with opposite parity and N odd, then N is a square and $M = 2^{k_0} \cdot$ (odd square).

Hygiene. Do not elevate "billion pairs found" to infinitude. Do not treat absence in catalogs as nonexistence proof.

Method (box)

1. **σ-sieve.** For $i = 1..$`SIGMA_LIMIT`, add i to $\sigma[j]$ for multiples j of i (`SIGMA_LIMIT = 3\cdot 10^6`).
2. **Amicable test.** For each $a \leq 10^6$, set $b = \sigma(a) - a$. Accept iff $b > a$ and $\sigma(b) - b = a$ (equivalently $\sigma(a) = \sigma(b) = a + b$).
3. **Flags.** Record parity of each member and $\gcd(a, b)$.
4. **Validation.** Compare counts to OEIS A066873.
5. **Label.** Finite catalog certification — **not** a proof of infinitude or of (non)existence of opposite-parity / coprime pairs.

Code: `experiments/band1/amicable_pairs/enumerate_amicable.py` → `results.json`.

Results

Catalog match

Quantity	Value
Bound (smaller $\leq$)	10^6
Pairs found	**42**
Pairs with smaller $< 10^5$	**13**
OEIS A066873 match	**Yes** ($a(5) = 13$, $a(6) = 42$)
Runtime	≈ 10.8 s

Parity / gcd ledger (within bound)

Flag	Count
Opposite parity	**0**
Coprime	**0**

Flag	Count
Both even	**34**
Both odd	**8**

All 42 pairs (smaller, larger)

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(220, 284), (1184, 1210), (2620, 2924), (5020, 5564), (6232, 6368),
(10744, 10856), (12285, 14595), (17296, 18416), (63020, 76084),
(66928, 66992), (67095, 71145), (69615, 87633), (79750, 88730),
(100485, 124155), (122265, 139815), (122368, 123152), (141664, 153176),
(142310, 168730), (171856, 176336), (176272, 180848), (185368, 203432),
(196724, 202444), (280540, 365084), (308620, 389924), (319550, 430402),
(356408, 399592), (437456, 455344), (469028, 486178), (503056, 514736),
(522405, 525915), (600392, 669688), (609928, 686072), (624184, 691256),
(635624, 712216), (643336, 652664), (667520, 696730), (726104, 796696),
(802725, 863835), (879712, 901424), (898216, 980984), (947835, 1125765),
(998104, 1043096)
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Both-odd pairs in bound (8)

Smaller	Larger	gcd
12285	14595	105
67095	71145	135
69615	87633	819
100485	124155	45
122265	139815	585
522405	525915	585
802725	863835	105
947835	1125765	135

All eight have gcd > 1 (as expected from the global empirical pattern).

Open questions (honest)

1. **Opposite-parity existence.** Does there exist an amicable pair with one even and one odd member? **OPEN.** Conditional structure known (D'Abramo); no example in catalogs through 10^{20+} / BOINC 10^{21}.
2. **Coprime existence.** Does there exist an amicable pair with $\gcd(a, b) = 1$? **OPEN.** No example known; all catalogued pairs share a common factor.
3. **Infinitude.** Are there infinitely many amicable pairs? **OPEN.** Do **not** claim; catalog size is not a proof.
4. **Odd–odd infinitude / density.** Odd–odd pairs exist, but structural density questions remain open beyond enumeration.

What this probe does not claim

- Infinitude of amicable pairs.
- Nonexistence (or existence) of opposite-parity or coprime pairs.
- Any Clay problem, ToE closure, or GATE-04 unlock.
- Exhaustive search beyond $B = 10^6$ on this box (literature catalogs go far further).

Deliverables

Artifact	Path
Lab	`experiments/band1/amicable_pairs/`
Enumerator	`enumerate_amicable.py`
Results	`results.json`
Lit ledger	`literature_ledger.json`
This note	`papers/2026-09-11_Band1_amicable_pairs_probe.md` (+ PDF)

References (selected)

1. OEIS A002025, A002046, A066873.
2. S. Chernykh, Amicable numbers list, https://sech.me/ap/ ; BOINC Amicable (search to 10^{21} finished 2026-04).
3. G. D'Abramo, On Amicable Numbers With Different Parity, arXiv:math/0501402.
4. H. J. J. te Riele, Computation of all the amicable pairs below 10^{10}, *Math. Comp.* **47** (1986).
5. Z. Chen, H. Tang, J. Zhan, Formalization of Amicable Numbers Theory, arXiv:2601.07444 (2026).
6. google-deepmind/formal-conjectures issues #2241 (opposite parity), #2244 (coprime).

GATE-04 LOCKED. No Clay. No ToE. Infinitude not claimed. Opposite-parity / coprime: **OPEN**.